

From Mimetic to Structured Initialization: Reproducing Attention Biases for Small-Data Vision Transformers

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GitHub: <https://github.com/Lu-Shen939/CS5782-Final-Report---vit-attention-initialization-reproduction>

1 Introduction

Vision Transformers (ViTs) can perform very well at scale, but they are often less data-efficient than convolutional neural networks when trained from scratch on small image datasets. CNNs contain locality and translation bias by design, while vanilla ViTs must learn spatial structure through attention. This project studies whether useful spatial bias can be injected only through initialization, without changing the ViT architecture.

We reproduce and connect two related ideas. Mimetic initialization observes that pretrained transformers contain structured attention-weight statistics, especially diagonal patterns, and initializes a new model to mimic those statistics. Structured initialization goes further by directly programming the initial attention maps to resemble local convolutional impulse filters. Our goal is to test whether these initialization methods improve CIFAR-10 and CIFAR-100 performance in full-data and low-data settings.

2 Chosen Result

The chosen result is the accuracy and attention-structure comparison among three ViT initializations: Default, Mimetic, and Structured. This combines the main claims of the two papers: Mimetic should improve training by copying pretrained-like self-attention statistics, while Structured should improve small-data learning by explicitly imposing local attention at initialization. We therefore reproduce both quantitative accuracy trends and qualitative attention-map structure. The quantitative result is summarized later in Figure 2, and the attention-map component is shown in Figure 3.

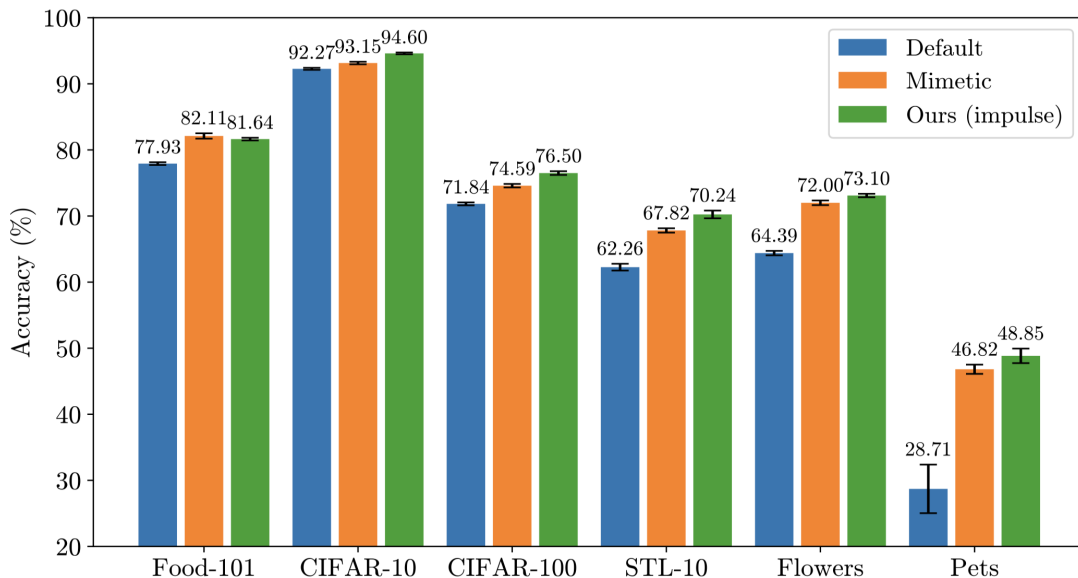


Figure 1: Original paper’s reported ViT-Tiny accuracy comparison across datasets (adapted from Zheng et al., 2025). This is the quantitative trend we use as the target for reproduction: Default, Mimetic, and Structured/Impulse initialization are compared under the same model family.

3 Methodology

We reimplemented the experiment in PyTorch using a ViT-Tiny style model from `timm`: 12 transformer layers, embedding dimension 192, 3 attention heads, image size 32×32 , and patch size 4. We evaluated CIFAR-10 and CIFAR-100 with full training data, plus 10% and 25% training-subset settings to test the small-data regime. The compared methods were default truncated normal initialization, Mimetic initialization, and Structured impulse initialization.

We used the official training recipe with AdamW, learning rate 2×10^{-3} , weight decay 0.05, cosine scheduling, 10 warmup epochs, RandAugment, Mixup, CutMix, label smoothing, random erasing, and stochastic depth. Full-data runs used batch size 1024 for 300 epochs on CIFAR-10 and 250 epochs on CIFAR-100; low-data runs used batch size 512 for 150 epochs on a Colab A100. Mimetic initializes attention projections toward pretrained-like products ($QK \approx I$, $VO \approx -I$), while Structured optimizes Q/K from positional embeddings so initial attention approximates sparse local impulse filters. We report best top-1 accuracy, training curves, seed stability, and trained patch-to-patch attention maps.

4 Results and Analysis

Figure 2 summarizes our reproduced results across six settings: full CIFAR-10/100, 10% CIFAR-10/100, and 25% CIFAR-10/100. Compared with the original paper trend in Figure 1, our absolute numbers are lower but the method ordering is similar: both initialization methods improve over the default ViT initialization, and Structured initialization is usually strongest. On full CIFAR-10, accuracy improves from 89.12% for Default to 91.99% for Mimetic and 92.47% for Structured. On full CIFAR-100, the same ordering appears: 67.95%, 72.67%, and 73.40%. This suggests that attention initialization is a useful way to inject inductive bias without changing the ViT architecture.

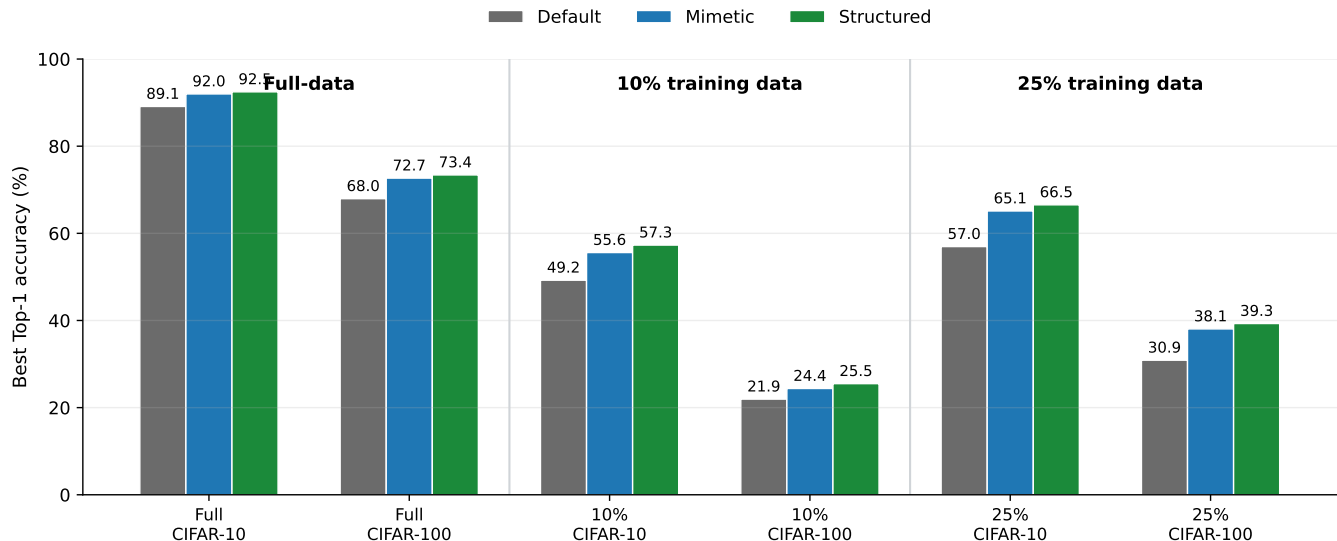
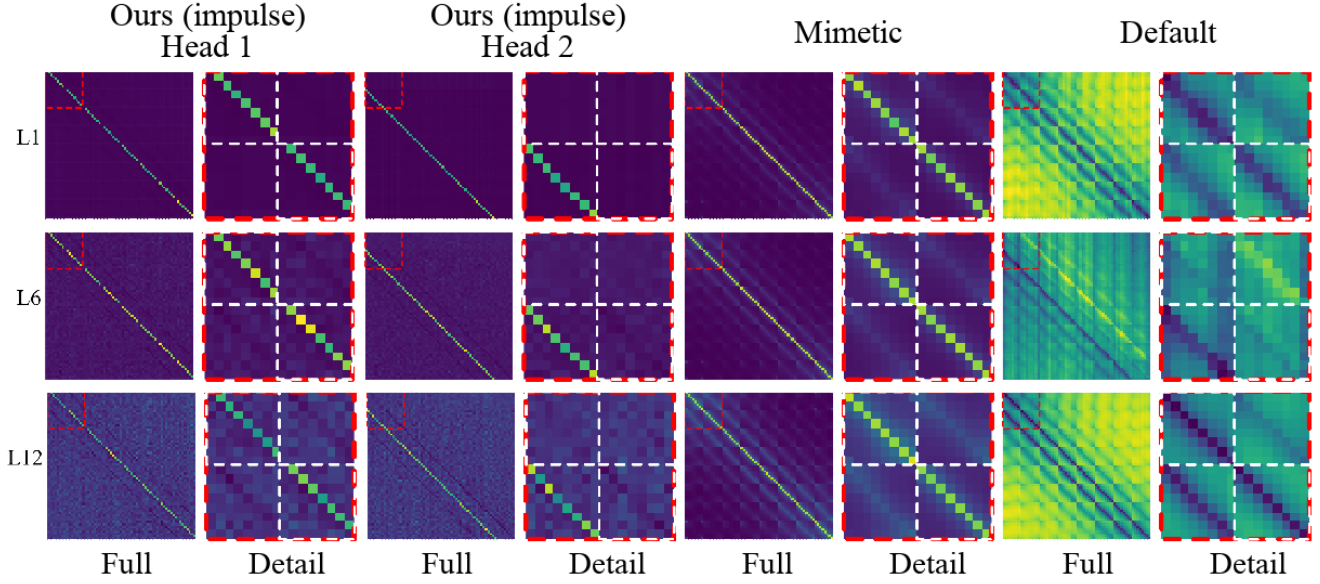


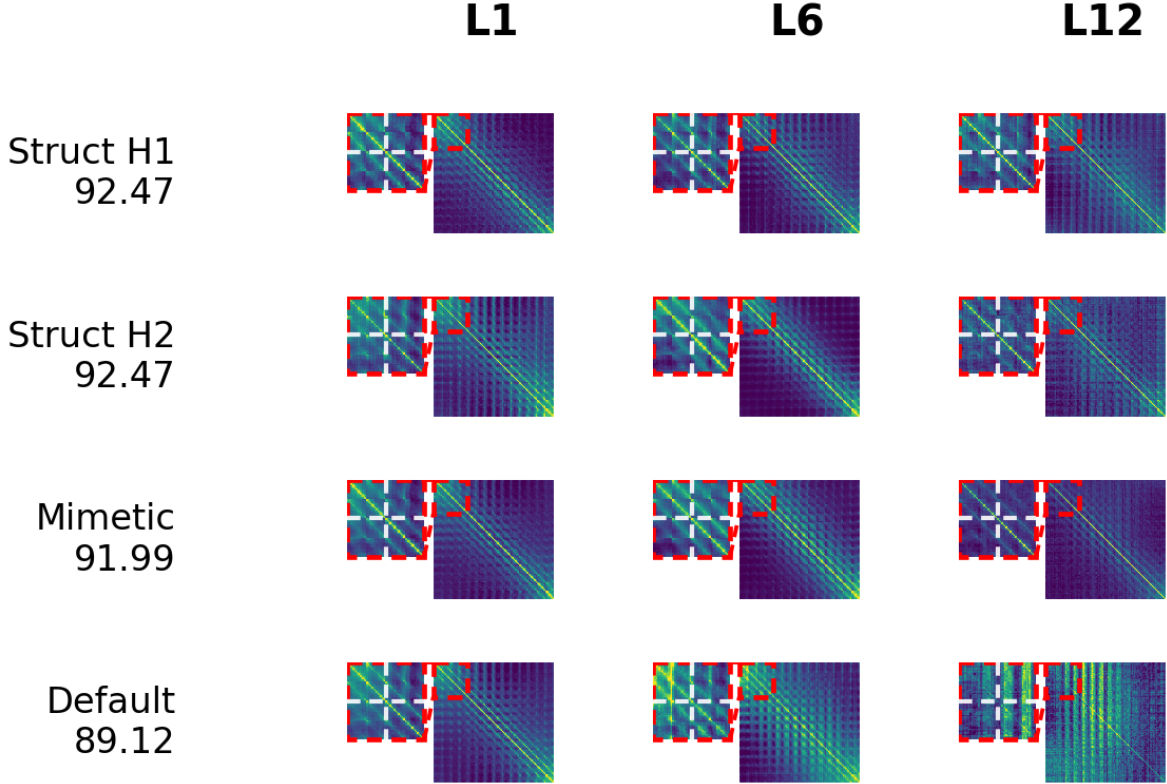
Figure 2: Best top-1 accuracy across full-data and low-data settings. Structured initialization gives the best result in most settings, while Mimetic also consistently improves over the default baseline.

The advantage is especially visible in the low-data experiments. With 10% of CIFAR-10, Mimetic improves over Default by 6.35 points, while Structured improves by 8.06 points. With 25% of CIFAR-10, the gains become 8.17 and 9.58 points. For CIFAR-100, Structured also improves over Default by 3.55 points at 10% data and 8.42 points at 25% data. These results support the small-data motivation: when less supervision is available, a spatial attention prior helps the model avoid learning locality completely from scratch.

Qualitatively, Mimetic helps by copying pretrained-like self-attention statistics, while Structured is more explicit: it initializes query/key attention to resemble local convolutional impulse filters. The trained attention maps in Figure 3 provide a visual check: Mimetic and Structured runs show clearer diagonal/local patterns than a purely default initialization. The L1/L6/L12 comparison also follows the original paper visualization, where the impulse initialization produces strong local diagonal attention.



(a) Original paper attention-map figure at L1, L6, and L12 (adapted from Zheng et al., 2025).



(b) Our trained CIFAR-10 seed-0 checkpoint attention maps.

Figure 3: Attention-map comparison. Both panels show patch-to-patch attention patterns at L1/L6/L12; brighter diagonal structures indicate stronger local patch attention.

Our absolute accuracies are still below the full results reported in the original papers. A likely reason is that our reproduction used a shorter few-hundred-epoch training budget, fewer seed/hyperparameter sweeps, and an independent reimplement. Therefore, the most important reproduced result is the relative pattern rather than exact matching: Default < Mimetic < Structured in most full-data and low-data settings.

Table 1: CIFAR-10 seed stability under a common 250-epoch budget.

Initialization	Seed 0	Seed 1	Seed 2	Mean $\pm$ Std
Default	88.20	87.72	87.44	87.79 ± 0.38
Mimetic	91.32	91.03	90.42	90.92 ± 0.46
Structured	91.85	90.90	89.92	90.89 ± 0.97

The seed study adds an important caveat. On CIFAR-10 at 250 epochs, Structured is best for seed 0, but has higher variance across seeds (0.97 std) than Mimetic (0.46 std) or Default (0.38 std). This suggests that the stronger structured locality prior can be more sensitive to the early training trajectory. When the prior aligns well with the data order and augmentation noise, it helps; when it does not, the model may need more training to adapt from local attention toward more task-specific attention patterns.

Figure 5 shows the corresponding training curves from the exported logs. Mimetic and Structured generally rise faster and end higher than Default, so their advantage appears during optimization rather than only at the final checkpoint.

5 Reflections

This project was successful in reproducing the main qualitative conclusion of the two papers: better attention initialization can improve small-data ViT training without modifying the architecture. The clearest result is that Mimetic initialization is a reliable improvement over the default baseline, and Structured initialization usually provides the largest gain, especially in low-data settings.

The main limitation is that our absolute accuracies are below the original reported results. This is expected for a course-scale reproduction because we used a smaller compute budget, fewer seeds, and a simplified training sweep. The seed study also shows that Structured initialization is not uniformly stable: its stronger locality bias can improve data efficiency, but may also increase sensitivity to the exact run. With more time, we would train for the full original schedule, sweep learning-rate and augmentation settings, and evaluate more seeds to separate method-level gains from run-to-run variance.

6 References

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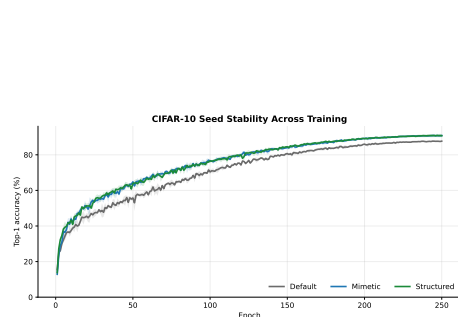


Figure 4: CIFAR-10 seed stability over seeds 0, 1, and 2.

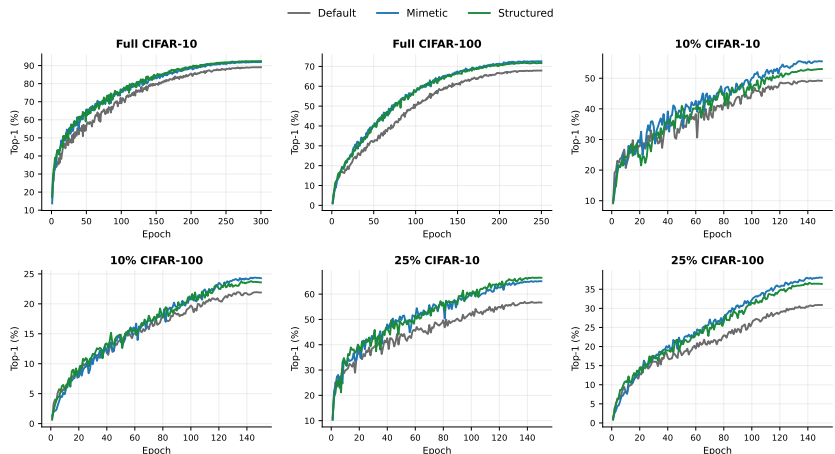


Figure 5: Training curves for the six evaluated settings.